

Factors Associated with Recurring Treatment Interruptions

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Background

Retention in care is the most significant gap in Jamaica's 90-90-90 cascade; interruption in HIV treatment and care is a primary contributor due to this indicator's stagnancy over the years. Our study assesses the factors associated with recurring treatment interruption cycles for PLHIV.

Introduction

The country currently uses the UNAIDS 90-90-90 targets to monitor the national HIV response, which states that 90% of PLHIV must be aware of their status, 90% of those aware of their status must be on treatment, and 90% of those on treatment must be virally suppressed.

Anecdotally, HIV care workers have lamented interruptions in treatment occurring among the same persons. Understanding the population-level engagement in treatment and factors associated with these patterns will be pivotal for developing strategies to combat the factors related to recurring interruptions in HIV treatment. A better understanding of the factors contributing to this phenomenon will lead to employing strategies to mitigate this, resulting in a long-term improvement in Jamaica's retention in care rate.

Objective

To determine the factors associated with recurring treatment interruptions, defined as patients with no clinic contact for ≥ 90 days after the last scheduled appointment on more than one occasion.

Methods

We conducted a cross-sectional cohort study of patients retained in HIV care in September 2023. Two cohorts were selected from data from the National Treatment Site Information System (TSIS 2.0) using random and purposive sampling.

Variables assessed:

- Sex at birth
- Age at HIV care registration
- Region of care
- Facility type
- Facility population size
- Marital status
- HIV transmission mode
- Suppression status based on the last available viral load test
- WHO HIV staging based on the first CD4 count upon linkage to care
- Current antiretroviral therapy line
- Test and start based on the time interval between the diagnosis date and the first ARV date by days
- Time in care based on the time interval between the first clinic date and the date of the data pull by year
- Average next appointment duration based on the duration of each next appointment given by months
- Average ARV prescription duration based on the duration of ARV given by month.

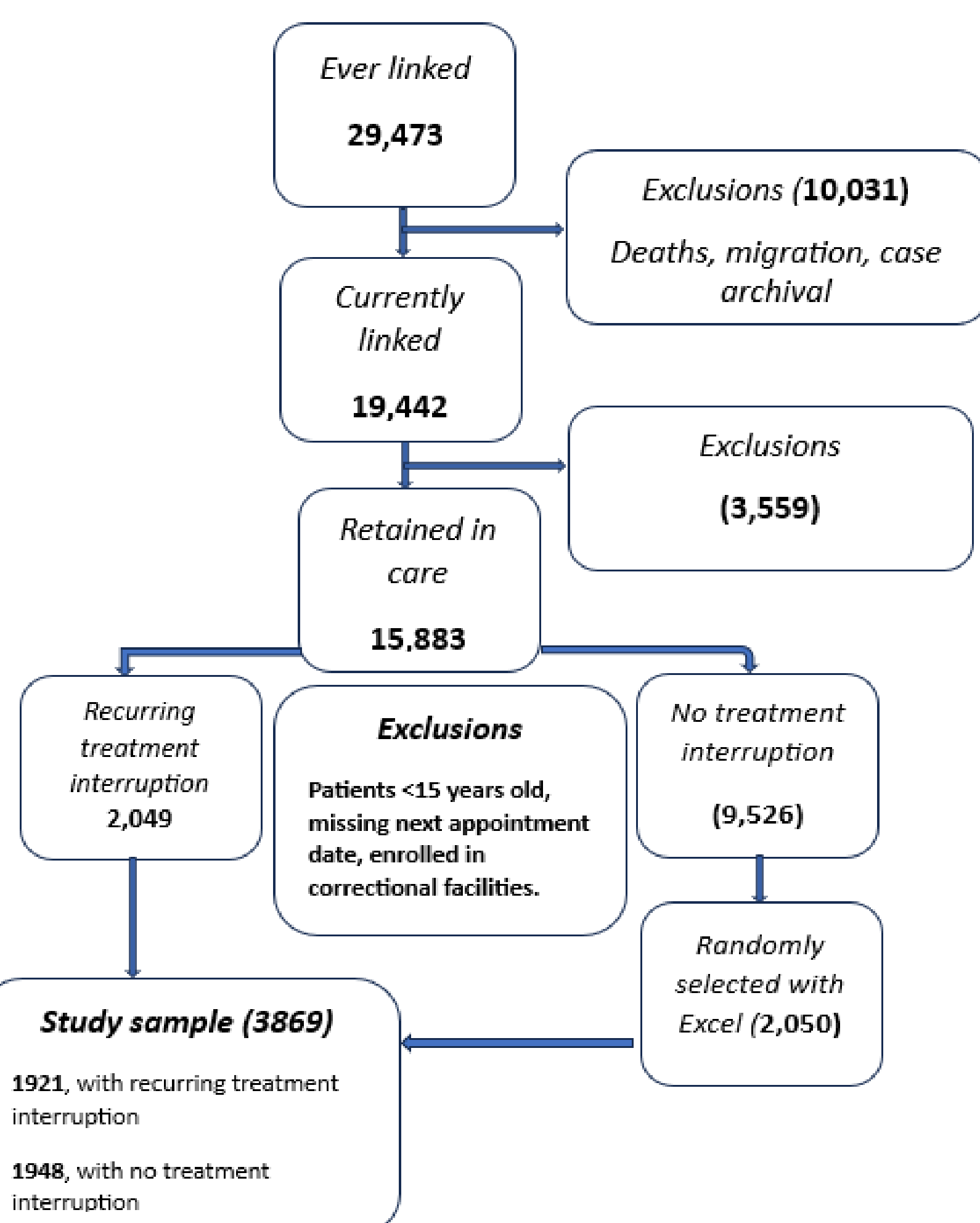


Figure showing the sample selection method for the study

Statistical Analysis

The Chi-square test and multivariable logistic regression were used to show an association of each demographic and clinical variable within the sample cohorts. The data was analyzed using IBM SPSS Version 26.

Results and Conclusions

Among 3,869 PLHIV retained in care in September 2023, 55.1% were female, and 44.9% were males, with a median age of 44; 50.3% of the sample showed recurring treatment interruption. Recurring treatment interruption was higher for PLHIV who registered for HIV between the ages of 0 and 14 years (79.9%), who accessed HIV treatment in NERHA (61.2%), who have been in care for 3 to 5 years (61.7%), and those who have an average appointment duration of 0-2 months (61.6%). The multivariable analysis showed recurring treatment interruption was likely for PLHIV with HIV treatment initiation at younger ages between 0 to 14 years (aOR=7.86 [4.46 – 13.72] vs. 60 and over),

being in care for 6 to 8 years

(aOR=1.66 [1.38 – 2.04] vs.

greater than nine years), and

having an average appointment

duration of 0 to 2 months

(aOR=37.93 [10.04 – 143.36] vs.

greater than eight months).

Table showing a multivariable analysis of the factors associated with cyclic interruption in care among people living with HIV in Jamaica, 2023

Variable	Multivariate OR [95%CI]	p-value p(<0.001)
Age at Registration		
0-14	7.86[4.46 – 13.72]	<0.001
15-29	3.36[2.19 – 5.15]	<0.001
30-59	2.15[1.414 – 3.27]	<0.001
60+	1[ref]	0
Time in care (years)		
<3	0.23[0.17 – 0.30]	<0.001
3-5	1.60[1.29 – 1.98]	<0.001
6-8	1.66[1.38 – 2.04]	<0.001
>=9	1[ref]	0
Test and start (days)		
Same day	1.00[0.80 – 1.27]	0.970
1-7	0.68[0.50 – 0.91]	0.009
8-14	0.94[0.65 – 1.36]	0.737
>14	1[ref]	0
Average ARV Duration (Months)		
0-2	0.38[0.21 – 0.65]	0.001
3-5	0.29[0.18 – 0.49]	<0.001
6-8	0.22[0.13 – 0.36]	<0.001
>8	1[ref]	0
Average Next Appt Duration (Months)		
0-2	37.93[10.04 – 143.36]	<0.001
3-5	15.62[4.20 – 58.07]	<0.001
6-8	3.57[0.07 – 13.14]	0.055
>8	1[ref]	0
Region		
Southeast	1.08[0.88 – 1.31]	0.475
Southern	1.27[0.98 – 1.64]	0.075
Northeast	1.46[1.12 – 1.91]	0.005
Western	1[ref]	0
Last Viral Load Result		
Virally suppressed	0.44[0.36 – 0.54]	<0.001
Unsuppressed	1[ref]	0

Other Key Findings

- HIV treatment initiation during paediatric to early adolescent years was associated with recurring treatment interruption.
- PLHIV patients with less than three years in care were less likely to show recurring treatment interruptions than those in care for three years or more.
- PLHIV, who indicated the presence of other comorbidities, were less likely to show recurring treatment interruption.
- PLHIV with an average appointment duration of 0-2 months were **37.9** times more likely to have recurring treatment interruption.

While current programme strategies for treatment care, and support for PLHIV have been proven effective, an overlooked phenomenon occurs in our treatment cascade. Shorter average appointment duration, longer time in care, and HIV initiation at a younger age are associated with recurring treatment interruption. A more diverse package of treatment options that satisfies programmatic obligations while prioritizing the patient's comfort and long-term satisfaction must be developed to mitigate recurring treatment interruptions.

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